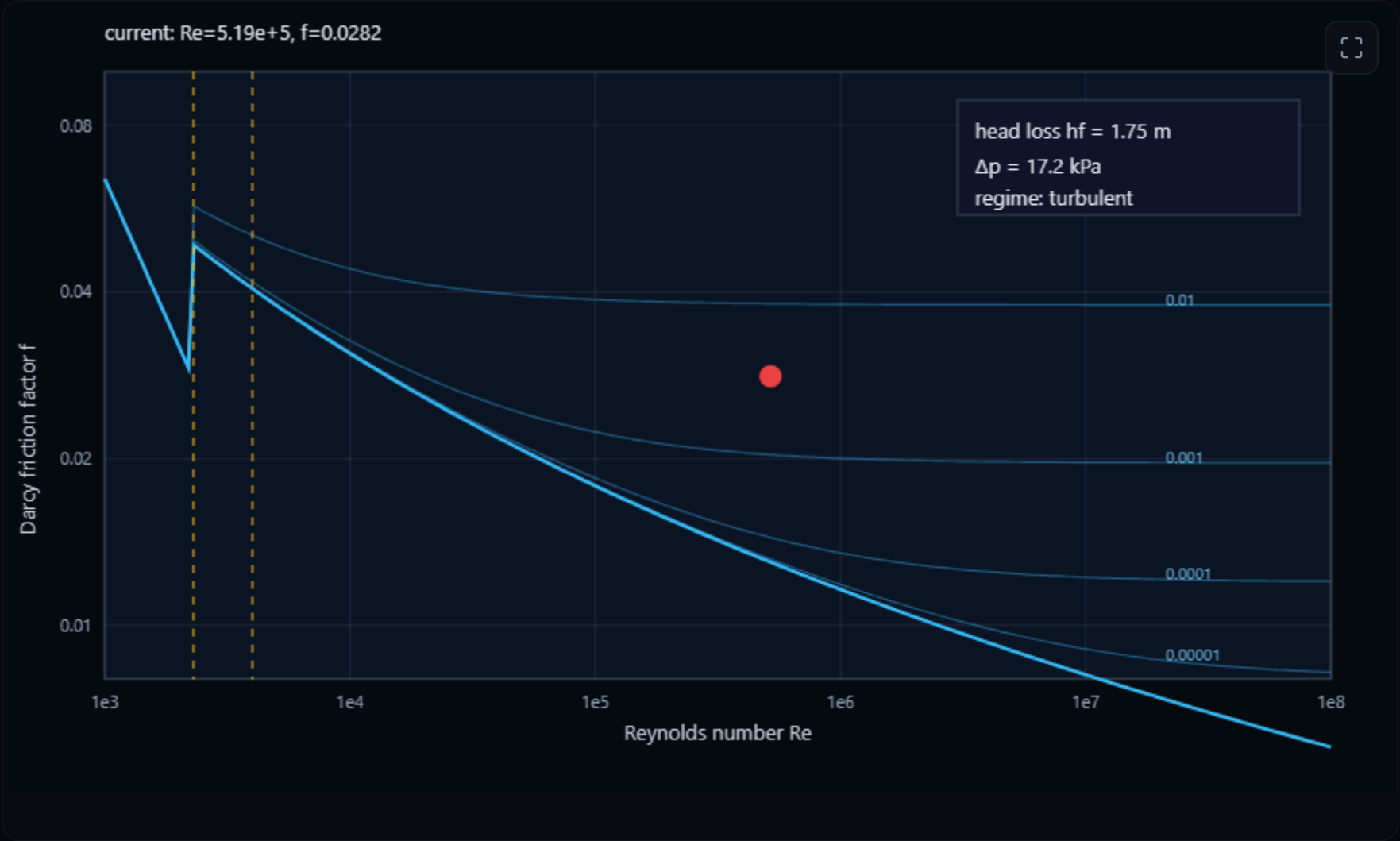


Pipe Friction & Moody Chart

Reynolds number, relative roughness, Swamee-Jain/Colebrook-style friction factor, and Darcy-Weisbach head loss.

This is a steady, fully developed pipe-flow model using Darcy friction factor. Minor losses, fittings, pumps, non-Newtonian fluids, compressibility, entrance effects, and two-phase flow are not included.

LIVE GRAPHS



PIPE AND FLOW

Pipe diameter D 390 mm

Pipe length L 268 m

Flow rate Q 159.2 L/s

Roughness ϵ 1.455 mm

FLUID

water

oil

Density ρ 998 kg/m³

Dynamic viscosity μ 1 mPa · s

MEASURED VALUES

Reynolds number 5.19e+5

Darcy friction factor 0.02817

Relative roughness ϵ/D 0.003731

Mean velocity 1.33 m/s

Head loss h_f	1.75 m
Pressure drop Δp	17.2 kPa

About this model

Pipe pressure loss in steady internal flow is commonly estimated with the Darcy-Weisbach equation, $h_f = f(L/D)V^2/(2g)$. The friction factor f depends on Reynolds number $Re = \rho V D / \mu$ and relative roughness ϵ/D . This simulator uses $f = 64/Re$ for laminar flow and the explicit Swamee-Jain approximation for turbulent flow, which tracks the Colebrook/Moody chart without iteration. The chart shows smooth and rough turbulent curves, transition markers, the current operating point, head loss, and pressure drop. It assumes steady, fully developed, incompressible, single-phase Newtonian flow in a straight circular pipe.

Who it's for: Fluid mechanics, hydraulics, HVAC, process piping, civil engineering, and mechanical engineering introductions.

KEY TERMS

- Moody chart
- Darcy-Weisbach
- Friction factor
- Reynolds number
- Relative roughness

How it works

Moody chart and Darcy-Weisbach pipe friction calculator: Reynolds number, relative roughness, Swamee-Jain friction factor, head loss, and pressure drop.

Key equations

$$Re = \rho V D / \mu, \quad h_f = f \left(\frac{L}{D} \right) \frac{V^2}{2g}$$

Swamee-Jain: $f = 0.25 / [\log_{10}(\epsilon / (3.7D) + 5.74 / Re^{0.9})]^2$; laminar $f = 64 / Re$

Frequently asked questions

Why use Darcy friction factor instead of Fanning?

The Moody chart and Darcy-Weisbach head-loss equation usually use the Darcy factor. The Fanning factor is one quarter of the Darcy factor, so mixing conventions causes a factor-of-four error.

What happens in the transition region?

Between roughly $Re = 2300$ and 4000 , flow can switch between laminar and turbulent depending on disturbances and inlet conditions. The simulator flags it rather than pretending the correlation is precise there.

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Heated-from-below layer: Ra vs $Ra_c \sim 1708$; schematic hex/roll pattern.